

Uday Wasu

BS in Data Science and Applications | B.Tech in Artificial Intelligence & Data Science

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CAREER OBJECTIVE

Artificial Intelligence and Data Science undergraduate with a strong interest in intelligent robotics and applied AI research. Experienced in computer vision, signal processing, and sensor-integrated systems for healthcare and autonomous applications. Seeking opportunities to work at the intersection of AI, robotics, and embedded intelligence.

CORE COMPETENCIES

- AI-Integrated Robotics and Intelligent Systems
- Embedded Systems Design and Hardware–Software Integration
- Computer Vision, NLP, and Sensor-Based Perception
- Control Systems and Autonomous Navigation (PID Control)
- Signal Processing and Data-Driven Engineering Solutions

EDUCATION

BS – Data Science and Applications

Indian Institute of Technology Madras

2023 – Present

B.Tech – Artificial Intelligence & Data Science

P. R. Pote Patil College of Engg. & Mgmt, Amravati

2023 – Present

PROJECTS

KARM – AI-Integrated Humanoid Robot for Elderly Assistance

- Designed and developed a humanoid robotic system for elderly assistance, focusing on mechanical design, electronics system architecture, and hardware component selection.
- Implemented embedded control and IoT communication, integrating health-monitoring sensors.
- Integrated logic-driven intelligence with basic NLP capabilities and vision-based perception using ESP32 and Raspberry Pi, enabling natural interaction and assistive functionalities.

QUATRO – Autonomous Quadruped Robot with Terrain-Adaptive PID Control

- Designed and built a four-legged robotic platform for industrial inspection and surveillance, implementing PID-based real-time control for smooth locomotion across rough, inclined, and obstacle-laden terrain.
- Integrated multiple sensors for environmental monitoring and object detection, ensuring autonomous decision-making.
- Developed the embedded control system on ESP32, robust hardware–software integration and reliable autonomous navigation.

Samarthya – AI-Based Interview Analysis Platform

- Developed an AI-driven interview evaluation platform analyzing speech rate, prosody, pitch, vocabulary usage, and answer accuracy to deliver actionable user feedback.
- Built the full-stack system using React, Express, PostgreSQL, and Python, deployed on a local college server and validated through real user testing.
- Enabled data-driven interview insights by integrating AI, NLP, and real-time analysis pipelines for performance assessment.

Additional Projects:

- Face Liveness Detection System using reflected light for authentication; Smart Garbage Segregation via computer vision.
- Early academic/IoT projects: chatbots with Flask & MongoDB, parking booking system and gesture-controlled robotic arm.

TECHNICAL SKILLS

Programming: Python, C

AI & Data: Computer Vision, NLP, Audio & Signal Analysis, Applied Machine Learning

Robotics & Embedded: ESP32, Raspberry Pi, Sensors & Actuators, PID Control, Autonomous Systems

Software & Web: React, Flask, Express, REST APIs, MongoDB, SQLite, PostgreSQL, Supabase

Electronics & Design: IoT Systems, Electronics System Design, Basic Mechanical Design, 3D Modeling (Blender)